diversified away, the remaining risk must be priced in equilibrium, leading to a value premium.

The Fama–French model itself is silent on why value carries a premium. In contrast, the CAPM provides a theory of how the market factor is priced and even determines the risk premium of the market (see equation (7.1) and also chapter 7). To go further, we need to delve into an economic reason for why the value premium exists.

In the pricing kernel formulation, any risk premium exists because it is compensation for losing money during bad times. The key is defining what those bad times are. Let's look at Figure 7.6 again. The bad times for value do not always line up with bad times for the economy. Certainly value did badly during the late 1970s and early 1980s when the economy was in and out of recession. We had a recession in the early 1990s when value also did badly, and the financial crisis in 2008 was unambiguously a bad time when value strategies posted losses. But the bull market of the late 1990s? The economy was booming, yet value stocks got killed. Rational stories of value have to specify their own definitions of bad times when value underperforms, so that value earns a premium on average.

Some factors to explain the value premium include investment growth, labor income risk, nondurable or "luxury" consumption, and housing risk. A special type of "long-run" consumption risk also has had some success in explaining the value premium.[26] During some of the bad times defined by these factors, the betas of value stocks increase. This causes value firms to be particularly risky.[27]

Firm Investment Risk

A key insight into the behavior of value and growth firms was made by Berk, Green, and Naik (1999). They build on a *real option* literature where a manager's role is to optimally exercise real investment options to increase firm value.[28] A firm in this context consists of assets in place plus a set of investment options that managers can choose (or not) to exercise. The CAPM is a linear model, and it turns out that CAPM does not fully work when there are option features (see also chapter 10). Berk, Green, and Naik show that managers optimally exercise investment options when market returns are low. These investment options are dynamically linked to book-to-market (and size) characteristics, giving rise to a value premium. Value firms are risky; their risk turns out to be the same conventional bad times risk as the CAPM or other macro-based factors. Be a value investor only if you can stomach losses on these firms during these bad times.

[26] For a labor income risk explanation, see Santos and Veronesi (2006). Parker and Julliard (2005) and Lustig and van Nieuwerburgh (2005) consider luxury consumption and housing risk, respectively. For "long-run" consumption risk, see, for example, Bansal, Dittmar, and Lundblad (2005).

[27] For time-varying betas of value stocks, see Lettau and Ludvigson (2001b), Petkova and Zhang (2005), Lewellen and Nagel (2006), Ang and Chen (2007), and Ang and Kristensen (2012).

[28] This literature was started by McDonald and Siegel (1985).